



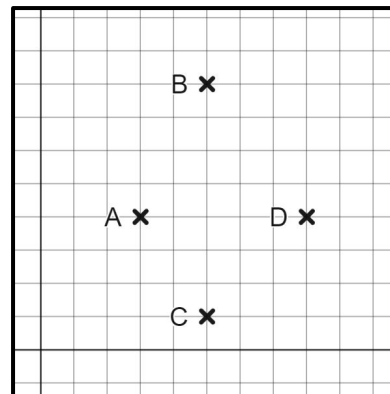
Rate of change from a coordinate pair

Task A: Reading the axes accurately to calculate gradient

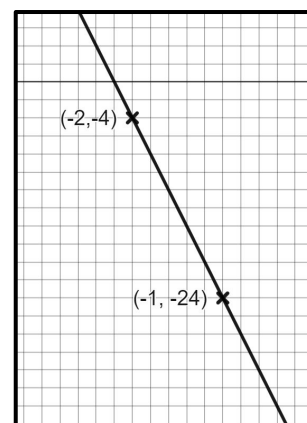
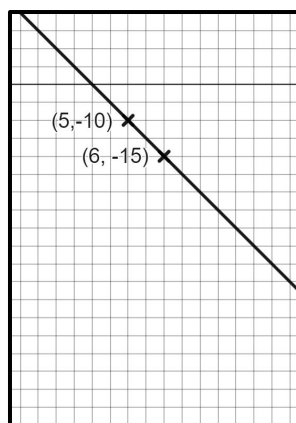
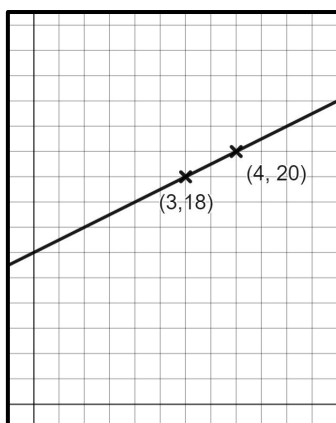
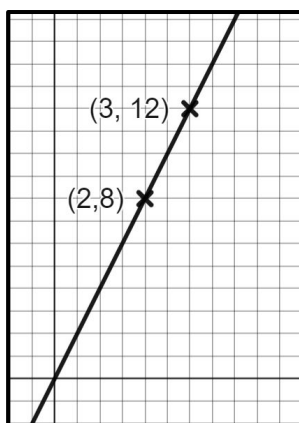
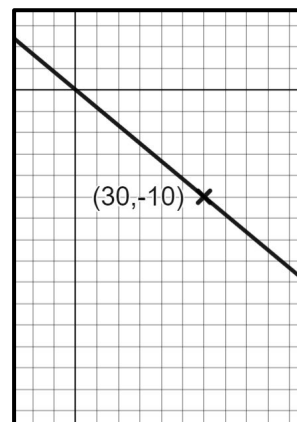
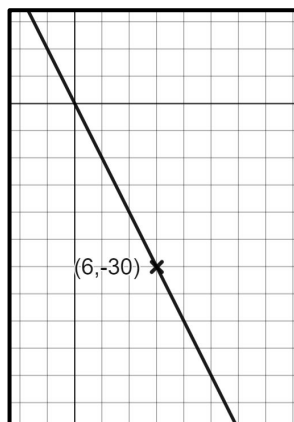
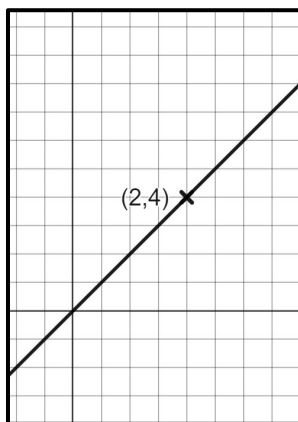
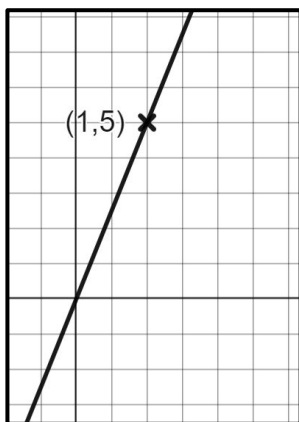
1) The points A, B, C and D have been plotted on the same axes.

Which is steeper

- a) A to B or C to D?
- b) A to C or B to D?
- c) A to B or A to C?
- d) Where could you move the coordinate, D, to make this shape a parallelogram?



2) Calculate the gradient of each line



Task B: Calculating gradient from coordinate pairs

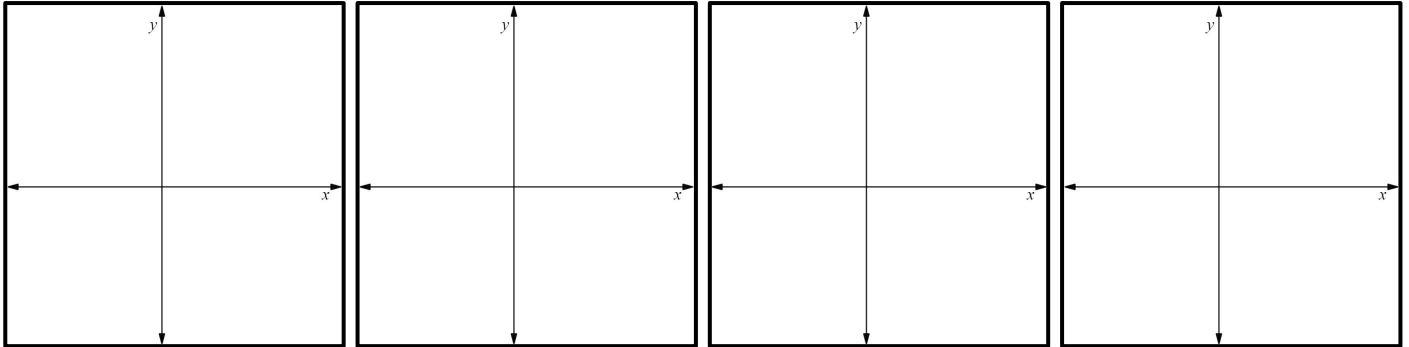
1) Find the gradient of the line joining the 2 coordinates given. You may use the axis to help

a) (7,8) and (4,2)

b) (-1,8) and (1,2)

c) (-1,-2) and (1,10)

d) (3,-2) and (-5,10)



2) In each question, decide which line will be steeper

a) The line joining (0,0) and (5,7) or the line joining (0,0) and (5,-4)

b) The line joining (0,0) and (3,4) or the line joining (0,0) and (6,7)

c) The line joining (1,2) and (2,5) or the line joining (5,1) and (6,5)

d) The line joining (-4,4) and (5,-4) or the line joining (2,-1) and (11, -8)

Task C: Coordinates that lie on a straight line

1) By testing the gradient. See whether each set of 3 points lie on a straight line.

a) $(1,2)$, $(2,6)$ and $(4,12)$

b) $(2,-8)$, $(6,-4)$ and $(13,3)$

c) $(0,9)$, $(2,5)$ and $(5,-1)$

d) $(-5, 0)$, $(-3,-4)$ and $(3,-8)$